1. Research Objective

This project focuses on Customer Analysis Segmentation for Zomato, a multinational restaurant aggregator and food delivery company. The objective is to answer three connected business questions: who are Zomato's most valuable customers, who are the registered users Zomato is failing to convert into buyers, and what does the overall customer base look like. Together these questions are intended to give the business a clear, actionable picture of where revenue is concentrated and where the largest untapped opportunity exists.

1.1 Why This Matters to Zomato's Business Model

Zomato's primary revenue stream is commission charged to restaurants on each completed order, supplemented by customer delivery fees, advertising, and subscription revenue (e.g. Zomato Pro). Because revenue is generated per order, customer ordering frequency and spend directly translate into company revenue. This means understanding which customers order often and spend more, and converting non-ordering registered users into active customers, are both directly tied to revenue growth.

2. Data Sources and Scope

Of the five tables provided (food, menu, orders, restaurant, users), this analysis will center on two:

- users — 100,000 rows. Contains demographic attributes: Age, Gender, Marital Status, Occupation, Monthly Income, Educational Qualifications, and Family size.
- orders — 150,281 rows. Contains order_date, sales_qty, sales_amount, currency, user_id, and r_id (restaurant ID).

The food and restaurant tables will be reviewed only briefly. They are mostly irrelevant to my analysis. Time-permitting, I may explore them further — for example, comparing vegetarian vs. non-vegetarian order patterns, or cuisine preference, across customer segments. The menu table is not relevant to this research area and will not be used.

2.1 Initial Data Exploration Findings

Preliminary exploration of the users and orders tables surfaced the following, which directly inform the approach below:

- The users table has zero missing values and zero duplicate user_id values, confirming it is reliable as the basis for a 1-to-many join with orders.
- Of 100,000 registered users, 77,929 (about 78%) have placed at least one order. The remaining 22,071 (about 22%) have never ordered.
- Order frequency is low and tightly compressed: customers placed between 1 and 9 orders over the data period, with an average of 1.93 orders per customer. 33,457 customers ordered exactly once.
- The order data covers October 2017 to June 2020, roughly two and a half years.

- sales_amount ranges from -1 (a data entry error, 2 occurrences) to 1,510,944 INR, with 1,609 orders valued at exactly zero. Outliers and these anomalies will be investigated and addressed during data cleaning, prior to scoring.
- The user base is demographically narrow and skewed young: ages 18-33 (average 24.6), 53% male, 69% single, 53% students, and 48% reporting no income. This base will shape how segments are interpreted.

3. Research Questions and Hypotheses

The plan is organized around three core questions, each with a corresponding hypothesis to be tested against the data.

Core Question 1 — Who are Zomato's most valuable customers?

- What demographic profile (age, gender, occupation, income, marital status) characterizes customers with high spend and high order frequency?
- Is customer value concentrated among a small group of customers, or spread broadly across the base?

Hypothesis 1: Employed customers with higher monthly income will show meaningfully higher total spend than students reporting no income.

Hypothesis 3: A relatively small share of customers (an estimated top 20%) will account for a disproportionately large share of total revenue, consistent with a Pareto-style distribution.

Core Question 2 — Who are the unconverted 22%, and how might they be converted?

- How does the demographic profile of users with zero orders differ from users who have ordered?
- Do registration timing patterns suggest some non-ordering users simply have not had time to order yet, versus users who have been registered for years without ordering?
- Are there distinct sub-groups within the non-ordering population that would call for different conversion approaches (for example, a first-order discount vs. general awareness messaging)?

Hypothesis 2: Non-ordering users will skew toward the youngest age range and the lowest income bracket compared to users who have placed orders.

Core Question 3 — What does the overall customer base look like?

- What is the complete demographic portrait of Zomato's registered customer base?
- Which identifiable segments represent the largest realistic growth opportunity for the business?

This question draws on the findings from Questions 1 and 2 to build a single, coherent narrative rather than introducing a separate hypothesis.

4. Methodology: Preparing the Data

Before any visualization or scoring takes place, the following data preparation steps will be completed. This sequencing — clean and verify first, build dashboard last — is intentional and will be followed in that order.

4.1 Data Cleaning

The anomalies identified during initial exploration were investigated prior to finalizing this plan, so that the cleaning approach below is evidence-based rather than assumed.

1. Negative sales_amount values (-1, 2 occurrences): both rows share an identical sales_qty (3) and order_date (2018-05-08). Comparing against 10 other orders with that same quantity and date, and against all 7,644 orders with sales_qty = 3 dataset-wide, normal values range from 5 to 12,620 INR. A value of -1 does not fall within or near this range and has no plausible business interpretation. Decision: exclude these 2 rows as data errors.
2. Zero-value orders (sales_amount = 0, 1,609 occurrences): checked for clustering by date and by user to distinguish a system error from a legitimate order type. These orders are spread across 488 distinct dates with no dominant date, and only 17 of 1,592 affected users have more than one such order. This broad, non-clustered pattern is inconsistent with a single system glitch or a small group of problem users, and is more consistent with legitimate free or promotional orders. Decision: retain these rows for frequency counts, but exclude them from monetary (spend) calculations, since a true 0 INR value would otherwise artificially lower average spend figures.
3. Maximum sales_qty (14,049): implied price per item (sales_amount ÷ sales_qty) is approximately 44.75 INR, in line with normal per-item food pricing. The next 9 highest-quantity orders show similarly plausible implied prices (42-74 INR per item). Decision: retain as a legitimate large bulk or corporate order rather than a data error.
4. Maximum sales_amount (1,510,944 INR): implied price per item is approximately 2,084 INR. The next 9 highest-amount orders show implied prices ranging 744-4,145 INR per item — high for typical food delivery, but not implausible for premium or large multi-item orders, and not the kind of absurd value (e.g., a misplaced decimal) that would clearly indicate an error. Decision: I'll retain with a documented caveat, as this could not be fully confirmed as legitimate or erroneous from the data alone.
5. Confirm date formatting and validate the full order_date range.
6. Convert all working ranges to Excel Tables to support reliable filtering, sorting, and formula referencing during analysis.
7. Retain all monetary values in INR (no currency conversion). Converting to USD would require an arbitrary exchange-rate choice, would not change any relative comparisons between customers, and does not match how the business itself reports.

4.2 Building the Customer-Level Dataset

Using a LEFT JOIN from users to orders on user_id (preserving all 100,000 users regardless of order history), the following will be calculated for each customer:

- Recency — days between the most recent order date in the dataset and each customer's own most recent order date. Lower values indicate more recent activity.
- Frequency — total count of orders placed by the customer (descriptive only; see 4.3).
- Monetary — sum of sales_amount across all of the customer's orders.
- Order status flag — whether the customer has ever ordered (Active) or never ordered (Non-Ordering), to separate the two populations cleanly for Question 2.

4.3 Scoring Approach: Modified RM Scoring (Recency and Monetary only)

A standard RFM model scores Recency, Frequency, and Monetary on a 1-5 scale using quintiles. After examining this dataset's Frequency distribution specifically, Frequency will be excluded from the formal score for two documented reasons:

- Range compression — order counts range only from 1 to 9 with an average of 1.93, which is too narrow to produce five meaningfully distinct quintile groups without misleadingly small breakpoint differences.
- Tenure blindness — raw order count does not distinguish a customer who placed 2 orders in their first week from one who placed 2 orders over two years; the two are not equally valuable, but a simple count treats them identically.

Recency and Monetary will each be scored 1-5 using quintiles (5 = most recent / highest spend). The two scores will be added for a combined score from 2-10. Order count will still be reported descriptively (e.g., percentage of customers who ordered only once vs. 4+ times) alongside the RM score, without being mathematically folded into it. This approach, and the reasoning behind excluding Frequency, will be stated explicitly in the final report.

4.4 Validation Checks

- Row counts before and after the join will be checked against expectations (100,000 customer rows after aggregation, regardless of order history).
- Duplicate user_id values will be re-confirmed as absent before finalizing the join.
- Total sales_amount after cleaning will be spot-checked against the raw total to confirm no unintended data loss.

5. Planned Visualizations

The dashboard will be built in Power BI and organized to tell a single connected story: best customers, missing customers, and the full picture. Each visualization maps to one of the three core questions.

Visualization	Purpose	Research Question
RM Score distribution (histogram)	Shows how customers spread across combined Recency + Monetary scores (2-10); identifies how many fall into top vs. bottom tiers.	Q1

Visualization	Purpose	Research Question
Revenue concentration chart (decile bar chart + KPI callout)	Tests Hypothesis 3 directly. A bar chart shows revenue share by customer decile (Top 10%, Next 10%, etc.); a KPI card next to it states the headline finding (e.g. “Top 20% of customers = X% of revenue”).	Q1
Demographic breakdown of top-tier vs. bottom-tier customers (bar charts)	Compares age, occupation, and income across high vs. low RM score customers; tests Hypothesis 1.	Q1
Active vs. Non-Ordering demographic comparison (side-by-side bars)	Directly tests Hypothesis 2 — compares age and income distribution between the two groups.	Q2
Registration timing for non-ordering users (line chart)	A line chart plotting count of non-ordering user registrations by month shows whether they registered recently (may simply not have ordered yet) or long ago (true non-converters).	Q2
Overall customer base demographic portrait (KPI tiles + bar charts)	KPI tiles for total users and total active users, plus one bar chart each for age distribution, gender, occupation, income bracket, and family size, giving a full-base view for context.	Q3
Order frequency distribution (descriptive bar chart)	Reports the share of one-time vs. repeat customers as plain descriptive context, separate from the RM score.	Q1 / Q3

6. Expected Findings and Recommendations Approach

The final report will present conclusions and business recommendations, not just descriptive findings, in line with the project's assessment criteria. The intended structure of the conclusions section is:

1. A clear profile of the highest-value customer segment, to inform retention and loyalty efforts.
2. A clear profile of the non-ordering segment, including whether it appears to be one group or several, to inform targeted conversion strategies (for example, first-order incentives for low-income, recently registered users rather than a single blanket offer to all 22,071 users).
3. An overall assessment of where the largest realistic revenue opportunity lies — protecting high-value customers, converting non-orderers, or both — based on what the data shows.

Limitations will be stated directly in the report, including the exclusion of Frequency from formal scoring, the compressed order-count range in this dataset, and the documented data-quality decisions described in Section 4.1 — particularly the high-value order that could not be fully confirmed as legitimate or erroneous.